$$Q = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} 3 \\ \beta \end{pmatrix}, \qquad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

$$J(x) = \frac{1}{2} x^{\mathsf{T}} Q x - b^{\mathsf{T}} x, \qquad x_2 \geq 0. \tag{1}$$

$$\nabla J(x) = Qx - b = 0 \tag{2}$$

$$2x_1 + x_2 = 3$$

$$x_1 + 2x_2 = \beta \tag{3}$$

$$x^{\mathrm{u}} = \begin{pmatrix} 2 - \beta/3 \\ 2\beta/3 - 1 \end{pmatrix}.$$

$$
\begin{aligned}
g(t) &= J\left(\frac{3-t}{2}, t\right) \\
&= \left(\frac{3-t}{2}\right)^2 + \frac{3-t}{2}t + t^2 \\
&\quad - 3\frac{3-t}{2} - \beta t \\
&= \frac{3}{4}t^2 + \left(\frac{3}{2} - \beta\right)t - \frac{9}{4}.
\end{aligned} \tag{4}
$$

$$t^* = \begin{cases} 0, & \text{if } \beta \leq \frac{3}{2}, \\ \dfrac{2\beta - 3}{3}, & \text{if } \beta > \frac{3}{2} \end{cases}, \qquad x^* = \begin{pmatrix} (3 - t^*)/2 \\ t^* \end{pmatrix}. \tag{5}$$

From (2) and (3), the unconstrained point is $x^{\mathrm{u}} = (2, -1)^{\mathsf{T}}$ when $\beta = 0$.
Equation (4) gives $g'(t) = \frac{3}{2}t + \frac{3}{2} > 0$ for $t \geq 0$.
Thus (5) yields

$$x^* = \begin{pmatrix} 3/2 \\ 0 \end{pmatrix}, \qquad J(x^*) = -\frac{9}{4}.$$